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SUPERSEDING
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MILITARY SPECIFICATION

FOR

CABINETS, MONITOR, CY-7359/FSS-9(V), CY-7360/FSS-9(V), AND CY-7361/FSS-9(V)

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of monitor cabinets with integral power supply and mounting facilities with interface connections for the status and alarm monitor modules.

1.2 Classification. Monitor cabinets shall be of the following types, as specified (see 6.2):

Type A - One Zone Monitor Cabinet, CY-7359/FSS-9(V).
Type B - Five Zone Monitor Cabinet, CY-7360/FSS-9(V).
Type C - Twenty-five Zone Monitor Cabinet, CY-7361/FSS-9(V).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

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MIL-P-116	- Preservation, Methods of.
MIL-C-52853	- Control Unit, Alarm Set: C-9421/FSS-9(V).
MIL-T-52877	- Transmission Group, Data: Transmitter Data: T-1257/FSS-9(V) and Receiver, Data: R-1861/FSS-9(V).
MIL-M-52878	- Monitor Module, Status ID-1921/FSS-9(V) and Monitor Module, Alarm, ID-1922/FSS-9(V).

STANDARDS

FEDERAL

FED-STD-101	- Test Procedures for Packaging Materials.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810 1/	- Environmental Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

1/ The "C" revision of MIL-STD-810 shall be used.

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DRAWINGS

ME

TA13220E3900	- Cabinet, Monitor, 1 Zone, CY-7359/FSS-9(V).
TA13220E3901	- Power Supply, 1 Zone Monitor.
TA13220E3902	- Signal Module Assembly.
TA13220E4000	- Cabinet, Monitor, 5 Zone, CY-7360/FSS-9(V).
TA13220E4001	- Power Supply, 5 Zone Monitor.
TA13220E4100	- Cabinet, Monitor, 25 Zone, CY-7361/FSS-9(V).
TA13220E4101	- Power Supply, 25 Zone Monitor.
D13220E3912	- Battery, 24 VDC, 5.0 AMP-HR.
D13220E3983	Printed Wiring Board, Logic.
D13220E3990	- Printed Wiring Board, Power Supply.
D13220E4008	- Battery, 24 VDC, 28 AMP-HR.
D13220D4017	- Printed Wiring Board, Power Supply.
D13220E4134	- Printed Wiring Board, Power Supply.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The 1 zone, 5 zone, and 25 zone monitor cabinets shall be in accordance with Top Assemblies TA13220E3900, TA13220E4000, and TA13220E4100, respectively, and as specified herein.

3.1.1 Major components. The monitor cabinets shall consist of the following major components:

a. Signal module assembly: TA13220E3902.
b. Primary power supply: TA13220E3901, TA13220E4001, and TA13220E4101 for the 1, 5, and 25 zone monitor cabinets respectively.

When specified (see 6.2), any of the major components listed above shall be furnished separately.

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3.1.2 Function. The monitor cabinets house the primary notification devices (alarm and status monitor modules) of the Joint-Services Interior Intrusion Detection System (J-SIIDS) system. The monitor cabinets interface directly with the alarm set, control unit, conforming to MIL-C-52853, via hard-wire interconnecting lines or a data transmission system via a single pair transmission line. The monitor cabinets provide primary and backup power to the associated Status and Alarm Monitor modules, conforming to MIL-M-52878, and the data receiver conforming to MIL-T-52877. Three types of monitor cabinets are provided to accommodate 1, 5, or 25 rack mounted, status and alarm monitor modules. Each monitor cabinet includes a regulated direct current (dc) power supply, a battery charging circuit, and a signal module. The primary input power is monitored and an indication of alternating current (ac) power presence is provided.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the monitor cabinet or major components when furnished separately. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of monitor cabinets from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish monitor cabinets conforming to this specification.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The monitor cabinets shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the monitor cabinets may be newly fabricated from recovered materials to the maximum extent practicable, provided the monitor cabinets produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the monitor cabinets.

3.4 Environmental conditions. The monitor cabinets and major components when furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.2), performance degradation, or malfunction when tested as specified in 4.5.2.5.

3.4.1 Operational.

a. Temperature range: -20 to +150 °F.
b. Humidity: up to 95 percent relative.

3.4.2 Nonoperational and storage.

a. Temperature range: -30 to +165 °F.
b. Humidity: up to 95 percent relative.
c. Shock: bench handling; equipment not operating (without battery).

3.5 Detailed technical requirements. The monitor cabinets shall perform as specified herein under any of the operating test conditions specified herein.

3.5.1 Wiring. Monitor cabinet internal wiring and wiring harnesses shall be in accordance with the wire list on the applicable top assembly drawing (TA13220E3900, TA13220E4000, TA13220E4100). Transmission line brackets containing lightning and EMI suppression components shall be in accordance with applicable top assembly drawings.

3.5.2 Signal module. The signal module shall contain an audible signaling device and ac power-status indicator lamps.

3.5.2.1 Audible signaling device. When actuated the signaling device shall sound at a frequency of 2900 $\pm$500 Hertz and an amplitude not less than 60 decibels (dB) reference 0.0002 microbars sound pressure level at a distance of 10 feet on axis (assuming "A weighting"). The signaling device shall be actuated whenever less than 1 volt dc (Vdc) is being applied to terminal TB1-1 of the signal module. It shall also be actuated whenever the ac power status indicator lamps are flashing (see 3.5.2.2). The audible signaling device shall be silent when both of the following conditions exist: a voltage of 4 to 5 Vdc is applied at TB1-1; the ac power status indicator lamps are continuously on or off (not flashing).

3.5.2.2 Monitor cabinet ac power status indicator lamps. When the monitor cabinet is operating on external 115 V ac power, the indicator lamps shall be

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continuously on. When the monitor unit is operating on standby power, 24 V batteries, the ac indicator lamps shall be continuously off. When the external ac power to the monitor cabinet is removed for a period of 3 ±0.5 seconds or when ac power is reapplied to the monitor cabinet after having failed, the lamps shall begin to flash at a frequency of 4 ±1/2 Hertz (Hz). When the ACKNOWLEDGE switch is momentarily depressed, the flashing lights shall go to continuously off if the ac power is not present at the time of acknowledge, or to continuously on if the ac power has been restored at the time of acknowledge. Actuation of the ACKNOWLEDGE switch shall at no time prohibit the receipt of any power status change from resulting in the operation described above. Actuation of the switch to the LAMP TEST position shall cause both lamps to be continuously lit while the switch is actuated.

3.5.3 Primary power supply. The primary supply shall operate from 110 to 125 volts, 48 to 62 Hz ac power. The power supply shall provide a 20 ±1 Vdc regulated output voltage, a nominal 22 volts ac (Vac) output voltage, and an unregulated dc voltage. This unregulated dc voltage shall provide a trickle charge and a fast charge current for maintenance of a standby battery supply. The trickle charge voltage shall be 27.6 ±0.3 Vdc at room ambient temperature of 77 °F with fully charged batteries as specified herein connected across TB2-2 and TB2-1 and with resistors of 80, 32, and 6.4 ohms across terminals TB3-2 and TB3-7 for the 1 zone, 5 zone, and 25 zone power supplies respectively. For ambient temperatures differing by more than 5 °F from 77 °F, the center value of the trickle charge voltage shall be in accordance with the equation V=29.3 - .022T where T is the ambient temperature in degrees F. The fast charge output voltage shall be 29.5 ±3.5 Vdc. The primary supply shall provide full rated 20 V load current while recharging the standby supply. The fast charge mode shall be initiated when the voltage at the battery terminals (TB2-2 to TB2-1) falls to 24.3 ±0.3 Vdc. The trickle charge mode shall be initiated when the battery terminal voltage rises 2 ±0.5 V above the fast charge cut-in threshold, and shall exist whenever the voltage is above the fast charge threshold provided fast charge operation has not been initiated. The power supply shall conform to the line regulation, load regulation, and ripple voltage requirements specified herein. A power off/on switch shall be provided which in the OFF position shall disconnect ac power and the standby battery power from the power supply circuitry. With the switch in the ON position, a loss of input ac power shall cause the power supply to automatically operate from the standby battery supply. With a battery (batteries) or equivalent standby supply capable of providing the required current at 24 Vdc, the power supply shall continue to provide the regulated 20 Vdc output specified and shall draw the rated 20 V current plus or minus 20 percent from the battery or equivalent standby supply. The twenty-five zone power supply shall have a thermal overload switch (thermostat) in the battery fast charge circuits which shall open on a temperature rise to +150 ±5 °F and close when the power supply temperature falls to +130 ±6 °F. When using resistors or open circuit instead of batteries across the battery terminals, voltages shall be as specified in the appendix test procedures.

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Power supply requirements

Maximum values required/allowed

Characteristic	1 Zone	5 Zone	25 Zone
Load Current			
a. 20 ±1 Vdc	0.25 ampere (A)	1.25 A	6.25 A
b. 27.6 Vdc (Trickle charge)1/	0.250 A	0.300 A	0.750 A
c. 26 to 33 Vdc (Fast charge)1/	2 A	4 A	8 A
d. 22 ±2 Vac2/	1.27 milliampere (mA)	1.27 mA	1.27 mA
Line regulation			
a. 20 Vdc	±0.5%	±0.5%	±0.5%
b. 27.6 Vdc	±1%	±1%	±1%
c. 22 Vac	±10%	±10%	±10%
Load regulation (No load to full load)			
a. 20 Vdc	±1%	±1%	±1%
b. 27.6 Vdc	-	-	-
c. 22 Vac	±5%	±5%	±5%
Ripple content (Periodic and random deviation)	400 millivolts (mV) Peak-to-peak (P-P), at 0.25 A	400 mV (P-P) at 1.25 A	400 mV (P-P) at 6.25 A

1/ Output terminal for b. and c. is the same. Voltage varies with load current.

2/ This voltage and currents apply for line input of 115 Vac with applied load not greater than 19 kilohms. Values for other allowed input voltages shall comply with the line regulation requirement stated herein.

3.5.4 Standby power. Standby power is supplied from 24-volt, sealed, rechargeable immobilized electrolyte batteries floating on the input to the 20-volt regulator. The batteries utilized are as follows: 1 zone: two 24V, 5 ampere-hour (A-H) in accordance with 13220E3912; 5 zone: one 24V, 28 A-H in accordance with 13220E4008; 25 zone: three 24V, 28 A-H in accordance with 13220E4008. Unless specified (see 6.2), batteries are not deliverable items.

3.6 Burn-in. Each monitor cabinet and major component when furnished separately shall be subjected to a 72-hour burn-in to reduce the number of infant mortality failures (see 4.5.2.4).

3.7 Reliability. The monitor cabinet shall demonstrate a mean-time-between-failure (MTBF) of 10,000 hours when tested as specified in 4.5.2.8.

3.8 Identification marking. The monitor cabinets and major components shall be identified in accordance with MIL-STD-130.

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3.9 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.7) for manufacturing and testing monitor cabinets and major components when furnished separately:

Item No.	Description	Identification	Quantity
1.	Stable base drawing of printed wiring board (PWB) logic.	13220E3983, Sh 2	1 1/
2.	Stable base drawing of PWB, power supply (1 Zone).	13220E3990, Sh 2	1 2/
3.	Stable base drawing of PWB, Power Supply (5 Zone).	13220E4017, Sh 1 & 2	1 (each sheet) 3/
4.	Stable base drawing of PWB, Power Supply (25 Zone).	13220E4134, Sh 2	1 4/

1/ Provided when procuring all monitor cabinets and signal modules.
2/ Provided when procuring 1 zone monitor cabinets and 1 zone power supplies.
3/ Provided when procuring 5 zone monitor cabinets and 5 zone power supplies.
4/ Provided when procuring 25 zone monitor cabinets and 25 zone power supplies.

3.10 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or adjoining parts, except for adjustment, and without selection for fit or performance.

3.11 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purpose intended.

3.11.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements speci-

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fied herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.3 Disassembly inspection. Failure of the first article to pass any examination or test shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Government-loaned property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspections. The monitor cabinets and major components when furnished separately shall be inspected as specified herein. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article monitor cabinets and major components when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests.

4.3.2.1 Complete monitor cabinets. Complete monitor cabinets shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection and for performing the disassembly inspection specified in 4.1.2.

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4.3.2.2 Major components furnished separately. Major components furnished separately shall be subjected to tests marked "X" in column 1 of tables II and III as applicable. Failure of any test shall be cause for rejection and for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, inspection level II. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination.

4.4.2.1 Individual. Each monitor cabinet and major component when furnished separately, shall be examined for the critical defect specified in 4.5.1. Presence of the defect shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major defects specified in 4.5.1. AQL shall be 0.4 percent defective.

4.4.3 Tests.

4.4.3.1 Individual.

4.4.3.1.1 Monitor cabinets. Each monitor cabinet shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.1.2 Major components furnished separately. Each major component when furnished separately shall be subjected to the tests marked "X" in column 2 of tables II and III as specified. Failure of any test shall be cause for rejection.

4.4.3.2 Samples.

4.4.3.2.1 Monitor cabinets. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I. AQL shall be 2.5 percent defective.

4.4.3.2.2 Major components furnished separately. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of tables II and III. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The monitor cabinet and major components when furnished separately shall be examined as specified herein for the following defects:

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Critical

1. Welding burn through.

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the monitor cabinets or major components when furnished separately.
106. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified.
107. Electromagnetic interference suppression components and networks not as specified.
108. Components missing or not as specified.
109. Assembly not correct.
110. Identification or special marking missing, incomplete or illegible.
111. Interchangeability not as specified.
112. Workmanship including wiring and soldering not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. After successful completion of functional testing, each monitor cabinet and major component when furnished separately shall be subjected to a 72-hour burn-in as specified in 4.5.2.4. Unless otherwise specified herein, the monitor cabinet and major components when furnished separately shall operate as specified herein without maintenance.

4.5.2.2 Test schedule.

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TABLE I. Test schedule for monitor cabinet.

First article	Quality Conformance		Tests	Requirement paragraph	Test paragraph
	Indiv	Sample			
1	2	3	4	5	6
X	X	X	Functional 1/	3.5.2, 3.5.3	4.5.2.6.1
X	X	-	Continuity	3.5.1	4.5.2.5
X	X	-	Burn-in 1/	3.6	4.5.2.4
X	-	X	Low temperature 1/	3.4.1a., 3.4.2a.	4.5.2.7.1
X	-	X	High temperature 1/	3.4.1a., 3.4.2a.	4.5.2.7.2
X	-	X	Humidity 1/	3.4.1b., 3.4.2b.	4.5.2.7.3
X	-	-	Shock	3.4.2c.	4.5.2.7.4
X	-	-	Reliability 1/	3.7	4.5.2.8

1/ Only the power supplies and signal modules of monitor cabinets need to be subjected to these tests. These components may be tested prior to installation in the monitor cabinet.

TABLE II. Test schedule for power supply.

First article	Quality Conformance		Tests	Requirement paragraph	Test paragraph
	Indiv	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.5.3	4.5.2.6.2
X	X	-	Burn-in	3.6	4.5.2.4
X	-	X	Low temperature	3.4.1a., .4.2a.	4.5.2.7.1
X	-	X	High temperature	3.4.1a., 3.4.2a.	4.5.2.7.2
X	-	X	Humidity	3.4.1b., 3.4.2b.	4.5.2.7.3
X	-	-	Shock	3.4.2c.	4.5.2.7.4

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TABLE III. Test schedule for signal module.

First article	Quality Conformance		Tests	Requirement paragraph	Test paragraph
	Indiv	Sample			
1	2	3	4	5	6
X	X	-	Functional	3.5.2	4.5.2.6.3
X	X	-	Burn-in	3.6	4.5.2.4
X	-	X	Low temperature	3.4.1a., 3.4.2a.	4.5.2.7.1
X	-	X	High temperature	3.4.1a., 3.4.2a.	4.5.2.7.2
X	-	X	Humidity	3.4.1b., 3.4.2b.	4.5.2.7.3
X	-	-	Shock	3.4.2c.	4.5.2.7.4

4.5.2.3 Test methods. Tests shall be conducted in accordance with the appendix test methods and as specified herein. Calibration of test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Burn-in. Each monitor cabinet and major component when furnished separately, shall be subjected to a 72-hour burn-in after successfully completing the functional test and prior to performing further tests. During burn-in, each unit shall be subjected to an abbreviated test as specified herein nine times per 24-hour period, but no test shall be initiated less than 45 minutes after completion of the previous test period. Except as specified during the test periods:

a. Each item under test shall be continuously energized by the primary power source specified;
b. Each power supply, whether as a component of a monitor cabinet or when furnished separately, shall have its 20 volt output deliver maximum rated current and its 27 volt output deliver the float charge current that results when resistor R3 of the applicable appendix test method is placed across this output in place of the battery.
c. Each signal module, whether as component of a monitor cabinet or when furnished separately, shall be operated such that its lamps are on steady.

Test to be performed 9 times daily shall be as follows (paragraph numbers referenced are those of applicable appendix test method):

Monitor cabinet: The following tests of appendix test method 2, 3, or 4 as applicable: output voltage levels (see 3.1); ripple and noise (see 3.4); standby power (see 3.5); and the tests of appendix test method 5, except the sound pressure level need not be measured. Any test steps requiring use of the batteries need not be performed.

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Power supply: output voltage levels (see 3.1); ripple and noise (see 3.4); standby power (see 3.5). Any test steps requiring use of batteries need not be performed.
Signal module: test specified in appendix test method 5, except sound pressure level need not be measured.

Noncompliance with the applicable portions of 3.5 shall constitute a failure. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order, failure and burn-in reports shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional 72-hour burn-in, after failure analysis and repair, and successful completion of the functional test.

4.5.2.5 Continuity. Each monitor cabinet shall be tested in accordance with appendix test method 1. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.6 Functional. Functional tests of the monitor cabinet and major components when furnished separately shall be in accordance with 4.5.2.6.1 through 4.5.2.6.3 as applicable.

4.5.2.6.1 Monitor cabinet. The 1, 5, and 25 zone monitor cabinets shall be subjected to the functional tests for the power supply in accordance with appendix test method 2, 3, or 4 as applicable, and the functional test for the signal module in accordance with appendix test method 5. To perform those tests, either the power supply and signal module shall be outside the monitor cabinet or the wires into terminals 2 and 3 of the signal module shall have been disconnected prior to connection into the respective test fixtures. Nonconformance to 3.5.2 and 3.5.3 shall constitute failure of this test.

4.5.2.6.2 Power supply. The power supply shall be tested in accordance with appendix test methods 2, 3, or 4 as applicable. Nonconformance to 3.5.3 shall constitute failure of this test.

4.5.2.6.3 Signal module. The signal module shall be tested in accordance with appendix test method 5. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.7 Environmental. The monitor cabinets and major components when furnished separately shall be subjected to the following series of tests specified in MIL-STD-810. Functional tests required before, during, or after environmental tests shall be in accordance with 4.5.2.6.

4.5.2.7.1 Low temperature. Conduct low temperature test method 502.1, procedure 1, storage temperature of -30 °F for not less than 48 hours, equipment operating temperature of -20 °F.

4.5.2.7.2 High temperature. Conduct high temperature test method 501.1, procedure II, except that the higher internal chamber temperature shall be +165 °F and the operating temperature shall be +150 °F.

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4.5.2.7.3 Humidity test. Conduct test method 507.1, procedure I with the following exceptions:

a. Step 5 shall be for 5 cycles (not less than 120 hours). Perform the functional test at the completion of each cycle.
b. During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
c. During step 4, the relative humidity shall be maintained at 92 percent or greater.
d. Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24-hour cycle. The doors of the monitor cabinet shall not be opened during measurements. The monitor cabinet may be removed from the environmental chamber for measurement. However, unsuccessful completion of measurements or unsuccessful return of the monitor cabinet to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.
e. The major components may be removed from the environmental chamber for testing. Testing shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. Nonconformance to the appropriate functional requirements for the major components when tested in accordance with the applicable test methods in the appendix and failure to return the major components to the test chamber within 1 hour and 15 minutes subsequent to initiation of testing shall constitute failure of the test.

4.5.2.7.4 Shock (bench handling). Conduct test method 516.2, procedure V, equipment not operating (without battery).

4.5.2.8 Reliability. Using an upper test MTBF (θ_o) of 10,000 hours and a lower test MTBF (θ_1) of 5,000 hours the monitor cabinet shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan VIIIC. Continue testing until either an "accept" or "reject" decision is reached. All test incidents will be scored by the Government. A failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

4.5.2.8.1 Test procedure. The tests shall be conducted under ambient laboratory conditions without temperature or humidity cycling. Equipment shall be energized at the primary power input specified herein. Load resistors R1 and R3 of the applicable test methods (2, 3, or 4) shall be placed across the 20 Vdc and 27 Vdc outputs respectively, except as required during the test periods described below. Operation shall be continuous throughout the test. Nine times each 24 hours, but not more often than once per hour, complete functional tests equivalent to those of appendix test methods 2, 3, or 4 (as appropriate) and 5 shall be performed, except tests to determine sound pressure level, line and load regulation, and fast charge cut-in and cut-out thresholds need be performed

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only once daily. The sound pressure level need not be measured provided no notable deviation from the normal level pressure level is heard. The test shall commence only after successful completion of functional testing and a burn-in period of 72 hours, as specified in 4.5.2.4. Failed indicator lamps shall be noted as to time failure is detected and shall be replaced. Such failures shall not be counted as relevant but any dependent failures caused thereby are counted as relevant. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. No adjustments shall be allowed during the test.

4.5.2.8.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article pack inspection.

4.6.1.1 Examination. The first article pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The level A or B first article pack of a complete monitor cabinet or major components furnished separately shall be subjected to the free fall drop test as described in FED-STD-101, method 5007.1, procedure E. Any damage to the monitor cabinets or major components furnished separately or shifting of the units within the pack shall constitute failure of the test and shall be cause for rejection.

4.6.2 Quality conformance inspection.

4.6.2.1 Unit of product. For the purpose of inspection, a finished pack of a complete monitor cabinet or major components when furnished separately shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. The AQL shall be 2.5 percent defective.

113.	Complete monitor cabinets not preserved as specified for levels A and B.
114.	Keys not placed in bag and secured as specified for levels A, B, and C.
115.	Major components furnished separately not preserved as specified for levels A, B, and C.
116.	Complete monitor cabinets not packed as specified for levels A, B, and C.

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the fusible downwards in a vertical direction with a steady continuous

117. Major components furnished separately not packed as specified for levels A, B, and C.
118. Strapping not as specified for level A.
119. Marking missing, illegible, or incorrect for levels A, B, and C.
120. Special marking not as specified.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack of complete monitor cabinets and major components when furnisned separately for examination and test within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied packaging and marking comply with the packaging requirements of this specification. Examination and tests shall be those specified in section 4 of this specification and shall be subject to surveillance and approval by the Government (see 6.8). The first article pack may be prepared utilizing either the first article or a production monitor cabinet or major components when furnished separately. When the first article is used, any packaging shall be removed by the contractor at no expense to the Government, to facilitate comparison between the first article and production units.

5.2 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The monitor cabinets and major components when furnished separately shall be preserved by one of the following procedures depending upon procurement of complete monitor cabinets or major components furnished separately.

5.2.1.1 Procurement of complete monitor cabinets. Each monitor cabinet shall be preserved in accordance with MIL-P-116, method IIb. The inner container shall conform to PPP-B-636, type CF, class domestic, variety and grade as applicable to the weight and size of the contents. The contents shall be cushioned and secured within the container to prevent movement or damage. The keys shall be placed in a cotton draw-string bag or heat-sealed bag and placed within the container. The outer container shall conform to PPP-B-636, type CF, class weather-resistant, variety and grade as applicable to the weight and size of the contents.

5.2.1.2 Procurement of major components when furnished separately. Major components when furnished separately shall be individually preserved in accordance with MIL-P-116, method IIe, using a close-fitting box conforming to PPP-B-566, or PPP-B-676, variety 1, style, type, and class optional. In addition, major components furnished separately that are subject to electrostatic or electromagnetic damage shall be protected as specified in MIL-P-116 (see 5.3.2).

5.2.2 Level B. Preservation shall be as specified in 5.2.1 for level A except method 1A-14 shall be used in place of method IIb, and method 1A-15 shall be used in place of method IIe.

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5.2.3 Level C. Each monitor cabinet with its keys shall be preserved together. Major components when furnished separately shall be individually preserved in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first using activity.

5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Level A. The monitor cabinets and major components when furnished separately, preserved as specified in 5.2, shall be packed by one of the following procedures depending upon procurement of complete monitor cabinets or procurement of major components furnished separately.

5.3.1.1 Procurement of complete monitor cabinets. Each monitor cabinet, preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed the weight and size limitations of the box. Closure and strapping of the box shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, size as applicable, finish B.

5.3.1.2 Procurement of major components when furnished separately. Components of like description, preserved as specified in 5.2, shall be packed together as specified in 5.3.1.2.

5.3.2 Level B. Monitor cabinets, and major components when furnished separately, preserved as specified in 5.2 shall be packed as specified in 5.3.1 for level A, except boxes shall be domestic type. As an alternate, type A monitor cabinets and major components when furnished separately may be packed in boxes conforming to PPP-B-636, V3c. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.3.3 Level C. Each monitor cabinet and major components when furnished separately, preserved as specified in 5.2, shall be packed in shipping containers that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.4 Marking. Marking for all levels shall be in accordance with MIL-STD-129 and the special marking specified in 5.4.1 and 5.4.2.

5.4.1 Special marking. Each shipping container shall be marked with project code labels in accordance with MIL-STD-129. The labels shall contain the project code "JID".

5.4.2 Sensitive electronic components. Major components when furnished separately, preserved as specified in 5.2.1.2 or 5.2.2 for components requiring electrostatic or electromagnetic protection, shall be additionally marked in accordance with MIL-STD-129 with a sensitive electronic device caution label and the following caution statement:

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CAUTION

OBSERVE PRECAUTIONS FOR HANDLING

ELECTROSTATIC SENSITIVE ITEMS

6. NOTES

6.1 Intended use. The monitor cabinets are for use as a primary notification device for Joint-Services Interior Instrusion Detection System (J-SIIDS).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of monitor cabinets required (see 1.2).
c. When major components are to be furnished separately and number to be furnished (see 3.1.1).
d. When a first article is required for inspection and approval, the number of units required, and the number required of monitor cabinets and major components furnished separately (see 3.2 and section 5).
e. When Government-loaned property is not required (see 3.9).
f. Degree of preservation and degree of packing required (see 5.2 and 5.3).
g. When first article pack is required (see 5.1).
h. Schedule, content, and format of burn-in failure reports (see 4.5.2.4).
i. Schedule for examination and testing of first article pack (see 5.1).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The contracting officer should include specific instructions in acquisition document regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first articles.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the test item is designed to alarm.

6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the monitor cabinet; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or operational requirements resulting in noncompliance of the monitor cabinet with the provisions of this specification.

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6.5 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of monitor cabinets.

6.7 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.9, when applicable (see 6.2).

6.8 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the monitor cabinets and separately procured components in accordance with this specification.

6.9 Subject term (key word) listing.

Alarm monitor units
Alarm systems
Electronic components
Intrusion detection system
Physical security
Printed circuit boards

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME
Navy - EC
Air Force - 99

Review activity:
DLA-GS

User activities:
Army - ER
Navy - SH, MC, CG

Preparing activity:
Army - ME

Project 6350-A148

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APPENDIX

TEST METHODS

FOR

MONITOR CABINETS AND MAJOR COMPONENTS

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether monitor cabinets and major components furnished separately under this specification conform to the requirements specified. Figures 1 through 4 contain data for fabrication of test fixtures and preparation of test set-ups which are required to facilitate testing of the monitor cabinets and major components furnished separately.

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APPENDIX

TEST METHOD 1

MONITOR CABINET

CONTINUITY TEST

1. TEST APPARATUS:

a. Multimeter - Simpson Model 260 or equivalent.

2. TEST PROCEDURE:

NOTE: Connections marked with asterisk (*) must be checked for continuity after installation of the power supply and signal module in the monitor cabinet and completion of all functional tests. Remaining connections may be checked at any time.

2.1 Using an ohmmeter, perform a continuity check between the monitor cabinet connector pins as follows:

TYPE A

From	To
TBB-1	TBA-5
TBB-2	TBA-6
* TB2-2	+ battery terminal connectors (2 wires)
* TB2-1	- battery terminal connectors (2 wires)

TYPE B

From	To
TBF-A	TBA-5
TBF-A	TBA-6
TBF-B	TBB-5
TBF-B	TBB-6
TBF-C	TBC-5
TBF-C	TBC-6
TBF-D	TBD-5
TBF-D	TBD-6
TBF-E	TBE-5
TBF-E	TBE-6
* TB2-1	+ battery terminal connector
* TB2-2	- battery terminal connector

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configure the equipment as in 2.2(a) and (b). (a) C

APPENDIX

TYPES A, B, AND C

From	To
* TB1-1 (sig-mod)	J1-14
* TB3-4	J1-9
* TB3-6	J1-10
J1-1	TBA-1I
J1-2	TBA-1L
J1-3	TBA-2
J1-4	TBA-3
* TB3-1	TB1-2 (sig-mod)
* TB1-3 (sig-mod)	J1-9
J1-5	TBA-4
J1-6	TBCC-2
J1-8	TBA-5
TBCC-1	Chassis ground
J1-16	TBA-6
J1-10	J1-5
* TB1-4 (sig-mod)	J1-10

TYPES B AND C

From	To
J1-9	J2-9
J1-9	J3-9
J1-9	J4-9
J1-9	J5-9
J1-10	J2-10
J1-10	J3-10
J1-10	J4-10
J1-10	J5-10
J1-14	J2-14
J1-14	J3-14
J1-14	J4-14
J1-14	J5-14
J2-1	TBB-1I
J2-2	TBB-1L
J2-3	TBB-2
J2-4	TBB-3
J2-5	TBB-4
J2-6	TBCC-3
J2-8	TBB-5
J2-16	TBB-6
J3-1	TBC-1I
J3-2	TBC-1L
J3-3	TBC-2
J3-4	TBC-3
J3-5	TBC-4
J3-6	TBCC-4
J3-8	TBC-5
J3-16	TBC-6
J4-1	TBD-1I
J4-2	TBD-1L
J4-3	TBD-2
J4-4	TBD-3
J4-5	TBD-4
J4-6	TBCC-5
J4-8	TBD-5
J4-16	TBD-6
J5-1	TBE-1I
J5-2	TBE-1L
J5-3	TBE-2
J5-4	TBE-3
J5-5	TBE-4
J5-6	TBCC-6
J5-8	TBE-5
J5-16	TBE-6
J2-10	J2-5
J3-10	J3-5
J4-10	J4-5
J5-10	J5-5

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APPENDIX

TYPE C

From	To
J5-9	J6-9
J5-9	J7-9
J5-9	J8-9
J5-9	J9-9
J5-9	J10-9
J5-9	J11-9
J5-9	J12-9
J5-9	J13-9
J5-9	J14-9
J5-9	J15-9
J5-9	J16-9
J5-9	J17-9
J5-9	J18-9
J5-9	J19-9
J5-9	J20-9
J5-9	J21-9
J5-9	J22-9
J5-9	J23-9
J5-9	J24-9
J5-9	J25-9
J5-10	J6-10
J5-10	J7-10
J5-10	J8-10
J5-10	J9-10
J5-10	J10-10
J5-10	J11-10
J5-10	J12-10
J5-10	J13-10
J5-10	J14-10
J5-10	J15-10
J5-10	J16-10
J5-10	J17-10
J5-10	J18-10
J5-10	J19-10
J5-10	J20-10
J5-10	J21-10
J5-10	J22-10
J5-10	J23-10
J5-10	J24-10
J5-10	J25-10
J5-14	J6-14
J5-14	J7-14
J5-14	J8-14
J5-14	J9-14
J5-14	J10-14
J5-14	J11-14
J5-14	J12-14
J5-14	J13-14
J5-14	J14-14
J5-14	J15-14
J5-14	J16-14
J5-14	J17-14
J5-14	J18-14
J5-14	J19-14
J5-14	J20-14
J5-14	J21-14
J5-14	J22-14
J5-14	J23-14
J5-14	J24-14
J5-14	J25-14
J6-1	TBF-1I
J6-2	TBF-1L
J6-3	TBF-2
J6-4	TBF-3
J6-5	TBF-4
J6-6	TBCC-7
J6-8	TBF-5
J6-16	TBF-6
J7-1	TBG-1I
J7-2	TBG-1L
J7-3	TBG-2
J7-4	TBG-3
J7-5	TBG-4
J7-6	TBCC-8
J7-8	TBG-5

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APPENDIX

TYPE C (Cont'd)

From	To
J7-16	TBG-6
J8-1	TBH-1I
J8-2	TBH-1L
J8-3	TBH-2
J8-4	TBH-3
J8-5	TBH-4
J8-6	TBCC-9
J8-8	TBH-5
J8-16	TBH-6
J9-1	TBJ-1I
J9-2	TBJ-1L
J9-3	TBJ-2
J9-4	TBJ-3
J9-5	TBJ-4
J9-6	TBCC-10
J9-8	TBJ-5
J9-16	TBJ-6
J10-1	TBK-1I
J10-2	TBK-1L
J10-3	TBK-2
J10-4	TBK-3
J10-5	TBK-4
J10-6	TBCC-11
J10-8	TBK-5
J10-16	TBK-6
J11-1	TBL-1I
J11-2	TBL-1L
J11-3	TBL-2
J11-4	TBL-3
J11-5	TBL-4
J11-6	TBCC-12
J11-8	TBL-5
J11-16	TBL-6
J12-1	TBM-1I
J12-2	TBM-1L
J12-3	TBM-2
J12-4	TBM-3
J12-5	TBM-4
J12-6	TBCC-13
J12-8	TBM-5
J12-16	TBM-6
J13-1	TBN-1I
J13-2	TBN-1L
J13-3	TBN-2
J13-4	TBN-3
J13-5	TBN-4
J13-6	TBCC-14
J13-8	TBN-5
J13-16	TBN-6
J14-1	TBP-1I
J14-2	TBP-1L
J14-3	TBP-2
J14-4	TBP-3
J14-5	TBP-4
J14-6	TBCC-15
J14-8	TBP-5
J14-16	TBP-6
J15-1	TBQ-1I
J15-2	TBQ-1L
J15-3	TBQ-2
J15-4	TBQ-3
J15-5	TBQ-4
J15-6	TBCC-16
J15-8	TBQ-5
J15-16	TBQ-6
J16-1	TBR-1I
J16-2	TBR-1L
J16-3	TBR-2
J16-4	TBR-3
J16-5	TBR-4
J16-6	TBCC-17
J16-8	TBR-5

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APPENDIX

TYPE C (Cont'd)

From	To
J16-16	TBR-6
J17-1	TBS-1I
J17-2	TBS-1L
J17-3	TBS-2
J17-4	TBS-3
J17-5	TBS-4
J17-6	TBCC-18
J17-8	TBS-5
J17-16	TBS-6
J18-1	TBT-1I
J18-2	TBT-1L
J18-3	TBT-2
J18-4	TBT-3
J18-5	TBT-4
J18-6	TBCC-19
J18-8	TBT-5
J18-16	TBT-6
J19-1	TBU-1I
J19-2	TBU-1L
J19-3	TBU-2
J19-4	TBU-3
J19-5	TBU-4
J19-6	TBCC-20
J19-8	TBU-5
J19-16	TBU-6
J20-1	TBV-1I
J20-2	TBV-1L
J20-3	TBV-2
J20-4	TBV-3
J20-5	TBV-4
J20-6	TBCC-21
J20-8	TBV-5
J20-16	TBV-6
J21-1	TBW-1I
J21-2	TBW-1L
J21-3	TBW-2
J21-4	TBW-3
J21-5	TBW-4
J21-6	TBCC-22
J21-8	TBW-5
J21-16	TBW-6
J22-1	TBY-1I
J22-2	TBY-1L
J22-3	TBY-2
J22-4	TBY-3
J22-5	TBY-4
J22-6	TBCC-23
J22-8	TBY-5
J22-16	TBY-6
J23-1	TBZ-1I
J23-2	TBZ-1L
J23-3	TBZ-2
J23-4	TBZ-3
J23-5	TBZ-4
J23-6	TBCC-24
J23-8	TBZ-5
J23-16	TBZ-6
J24-1	TBAA-1I
J24-2	TBAA-1L
J24-3	TBAA-2
J24-4	TBAA-3
J24-5	TBAA-4
J24-6	TBCC-25
J24-8	TBAA-5
J24-16	TBAA-6
J25-1	TBBB-1I
J25-2	TBBB-1L
J25-3	TBBB-2
J25-4	TBBB-3
J25-5	TBBB-4
J25-6	TBCC-26
J25-8	TBBB-5
J25-16	TBBB-6

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APPENDIX

TYPE C (Cont'd)

From	To
J6-10	J6-5
J7-10	J7-5
J8-10	J8-5
J9-10	J9-5
J10-10	J10-5
J11-10	J11-5
J12-10	J12-5
J13-10	J13-5
J14-10	J14-5
J15-10	J15-5
J16-10	J16-5
J17-10	J17-5
J18-10	J18-5
J19-10	J19-5
J20-10	J20-5
J21-10	J21-5
J22-10	J22-5
J23-10	J23-5
J24-10	J24-5
J25-10	J25-5
TBDD-A	TBA-5
TBDD-A	TBA-6
TBDD-B	TBB-5
TBDD-B	TBB-6
TBDD-C	TBC-5
TBDD-C	TBC-6
TBDD-D	TBD-5
TBDD-D	TBD-6
TBDD-E	TBE-5
TBDD-E	TBE-6
TBEE-F	TBF-5
TBEE-F	TBF-6
TBEE-G	TBG-5
TBEE-G	TBG-6
TBEE-H	TBH-5
TBEE-H	TBH-6
TBEE-J	TBJ-5
TBEE-J	TBJ-6
TBEE-K	TBK-5
TBEE-K	TBK-6
TBFF-L	TBL-5
TBFF-L	TBL-6
TBFF-M	TBM-5
TBFF-M	TBM-6
TBFF-N	TBN-5
TBFF-N	TBN-6
TBFF-P	TBP-5
TBFF-P	TBP-6
TBFF-R	TBR-5
TBFF-R	TBR-6
TBGG-S	TBS-5
TBGG-S	TBS-6
TBGG-T	TBT-5
TBGG-T	TBT-6
TBGG-U	TBU-5
TBGG-U	TBU-6
TBGG-V	TBV-5
TBGG-V	TBV-6
TBGG-W	TBW-5
TBGG-W	TBW-6
TBHH-X	TBX-5
TBHH-X	TBX-6
TBHH-Y	TBY-5
TBHH-Y	TBY-6
TBHH-Z	TBZ-5
TBHH-Z	TBZ-6
TBHH-AA	TBAA-5
TBHH-AA	TBAA-6
TBHH-BB	TBBB-5
TBHH-BB	TBBB-6
TB4-1	+ Ext. dc pwr. terminal
TB4-2	- Ext. dc pwr. terminal
*TB2-1	- battery terminal connectors (3 total)
*TB2-2	+ battery terminal connectors (3 total)

2.2 Using an ohmmeter, verify that a dc open (high impedance) exists from chassis ground to terminals 5 and 6 of the following terminal boards:

1 zone: TBA
5 zone: TBA through TBE
25 zone: TBA through TBZ (except no TBI or TBO), TBAA and TBBB

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APPENDIX

TEST METHOD 2

1 ZONE POWER SUPPLY FUNCTIONAL TESTS

1. TEST APPARATUS

a.	Oscilloscope (Scope)	- Tektronix Model 5440 with type 5A48 plug-in or equivalent.
b.	Voltmeter, dc (M3, M6)	- 3-1/2 digit or greater, minimum accuracy 0.25% of reading + 1 digit
c.	Ammeter, dc (M5)	- 3-1/2 digit or greater, minimum accuracy 0.75% of reading + 1 digit.
d.	Voltmeter, ac (M1, M7)	- 3-1/2 digit or greater, minimum accuracy 1% of reading + 1 digit.
e.	Ammeter, ac (M4)	- 3-1/2 digit or greater, minimum accuracy 1.5% of reading + 1 digit.
f.	Ammeter, dc (M2)	- 3-1/2 digit or greater, minimum accuracy 1.5% of reading + 1 digit.
g.	Autotransformer (VR1)	- General Radio Model W20M or equivalent.
h.	SPST switches (S1-S8)	- Leviton Model 19916 or equivalent.
i.	Resistor Decade (R5)	- Clarostat Model 240C or equivalent.
j.	Resistor, 80 ohm (R1)	- Ohmite/Dividohm, 100 ohm, adjustable, 12 watts, P/N 1012 or equivalent.
k.	Resistor, 160 ohm (R2)	- Ohmite/Dividohm, 200 ohm, adjustable, 12 watts, P/N 1014 or equivalent.
l.	Resistor, 20 kilohm (R6)	- Allen/Bradley/Type GB, 20K Ω, ±5%, 1 watt or equivalent.
m.	Resistor, 270 ohm (R3)	- Ohmite/Dividohm, 300 ohm, adjustable 12 watts, P/N 1016 or equivalent.
n.	Resistor, 15 ohm (R4)	- Ohmite/Dividohm, 25 ohm, adjustable, 100 watts, P/N 0958 or equivalent.
o.	Battery, 24V, 5A-H (B1, B2)	- Eagle-Picher, CF2495 (no substitute).
p.	Test fixture	- (See figure 1).

NOTE: Power supply, such as Hewlett-Packard Model 6291A or equivalent, is required for calibrating test fixture resistor only.

2. TEST PREPARATION

2.1 When this test is being performed as part of the monitor cabinet functional test, either completely remove the power supply from the cabinet prior to performing the test procedure, or disconnect the wires going into terminals 2 and 3 of the signal module terminal board. Care should be taken to insure these wires do not make contact with any other wires.

3. TEST PROCEDURE

3.1 Output voltage levels.

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APPENDIX

a. Connect the power supply under test in the test fixture as shown in figure 1. Set the power supply under test power switch in the OFF position and apply 115 Vac, 60 Hz line power to the autotransformer (VR1).
b. Open S1 through S8.
c. Adjust the VR1 for 115 Vac on M1.
d. Close S3.
e. Place the power supply under test power switch in the ON position. Allow at least 15 minutes warm-up before proceeding with test.
f. The following output voltage levels shall be observed:

Output	Location	Meter	Voltage
1	TB3-3 to TB3-5	M6	20 ±1 Vdc
2	TB2-2 to TB2-1	M3	See below
3	TB3-1 to TB3-6	M7	22 ±2 Vac

Output 2: 28 ±0.3 Vdc at 76 ±8 °F, 30.5 ±1.0 Vdc at -20 °F, 26.5 ±0.5 Vdc at +150 °F.

g. The output voltages must be within the above levels to be acceptable.

3.2 Line regulation.

a. Close S3, S4, and S5, and open S1, S2, S6, S7, and S8.
b. Place the power supply under test power switch on the ON position.
c. Adjust the VR1 for a 115 Vac reading on M1.
d. Observe the following output current levels:

Output	Location	Meter	Current
1	TB3-2 to TB3-7	M5	0.25 A ±5%
2	TB2-2 to TB2-1	M2	0 A
3	TB3-1 to TB3-7	M4	1.1 mA ac ±15%

e. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-3 to TB3-5	M6
2	TB2-2 to TB2-1	M3
3	TB3-1 to TB3-6	M7

f. Adjust the VR1 for 125 Vac.
g. Repeat step e.
h. Adjust the VR1 for 110 Vac.
i. Repeat step e.
j. For acceptable line regulation, the output voltage measurements taken in steps g. and i. shall not differ from those taken in step e. by more than the following:

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1. Output 1 ±0.5%.
2. Output 2 ±1%.
3. Output 3 ±10%.

3.3 Load regulation.

a. Close S3 and S7 and open S1, S2, S4, S5, S6, and S8.
b. Adjust VR1 for 115 Vac as indicated on M1.
c. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-3 to TB3-5	M6
3	TB3-1 to TB3-6	M7

d. Close S4 and S5. Repeat the voltage readings as in step c. M5 shall indicate 0.25 ±.013A and M4 shall indicate 1.100 ±0.170 mA ac.
e. Open S4 and close S6. Repeat the voltage readings as in step c. M5 shall indicate 0.125 ±.007A and M4 shall indicate 1.100 ±0.170 mA ac.
f. For acceptable load regulation, the output voltage measurements taken in steps d. and e. shall not differ from those taken in step c. by more the following:

1. Output 1 ±1%.
2. Output 3 ±5%.

3.4 Ripple and noise (periodic and random deviations).

a. Open S1, S2, S6, and S8 and close S3, S4, S5, and S7.
b. Measure the ouput ac P-P voltage at output 1 (TB3-3 to TB3-5) with the scope. The lead lengths from the scope to TB3 should be as short as possible.
c. For acceptable ripple and noise content, the ac P-P voltage shall be less than 400 mV.

3.5 Standby power.

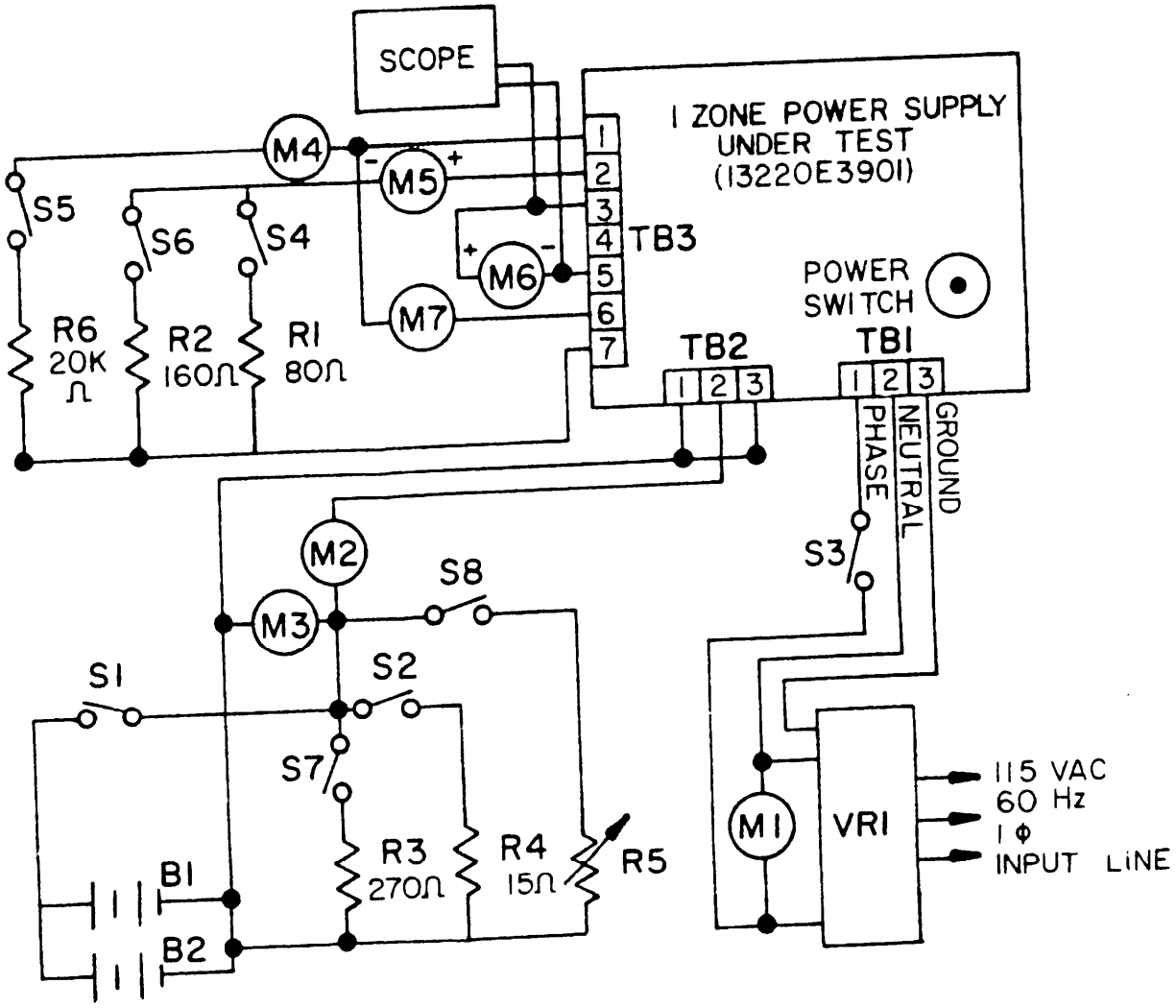
NOTE: Steps a., b., c., m., n., and o. shall not be performed during burn-in, high temperature, and low temperature tests.

a. Close S1, S3, S4, and S5 and open S2, S6, S7, and S8.
b. Observe current on M2. Current shall be less than 20 mA. If not, wait until current is less than 20 mA before continuing. If current will not become less than 20 mA, replace batteries with fresh set.
c. When current measured in step b. above is less than 20 mA, observe voltage on M3. Voltage shall be 27.6 ±0.3 Vdc at room ambient temperature of 77 ±5 °F. When ambient temperature is outside this range during "room temperature" testing or at 86 °F humidity test, the voltage shall be in accordance with the equation V=29.3 -.022T, where T is the temperature in degrees F.

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d. Close S3, S4, S5, and S7 and open S1, S2, S6, and S8.
e. Observe voltage on M3. Voltage shall be 27.0 ±0.5 Vdc at room ambient temperature of 77 ±5 °F, 29.6 ±0.6 Vdc at chamber temperature of -20 °F, and 25.8 ±0.6 Vdc at chamber temperature of 150 °F.
f. Remove scope probes from TB3 and place them across TB2-2 (positive) and TB2-1. Adjust vertical scale on scope to be able to read from 15 Vdc to 33 Vdc with best feasible resolution.
g. Open S7 and close S2. The voltage on scope shall fluctuate between 29.5 ±3.5 and 17 ±2 Vdc. Proceed to step h. within 15 seconds.
h. Open S2 and close S7. Voltage on M3 shall be within 0.2 Vdc of voltage measure in step e.
i. Adjust R5 for 270 ohms, then open S7 and close S8. Voltage as read on M3 shall be as in step h.
j. Slowly reduce resistance of R5 while observing voltage on M3 and scope. Note voltage on M3 just before the voltage begins to fluctuate. Voltage just prior to fluctuation shall be 24.3 ±0.3 Vdc at temperature of 77 ±10 °F, and shall be no less than 23 Vdc at -20 °F and no greater than 25 Vdc at ±150 °F. Proceed to step k within 15 seconds.
k. Increase resistance of R5 until voltage becomes steady. Voltage just prior to this change shall be 2 ±0.5 Vdc greater than that measured in step j.
l. Open S8 and close S1.
m. Open S3.
n. Observe voltage on meter M6 and current through meter M2. Voltage shall be 20 ±1 Vdc and current shall be 250 ±50 mA.
o. Turn off monitor cabinet power, open all switches S1 through S8, and restore wiring to original configuration, if applicable.

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CALIBRATION PROCEDURE

USING REGULATED POWER SUPPLY IN PLACE OF 13220E3901 POWER SUPPLY, ADJUST VARIABLE RESISTORS SUCCESSIVELY TO OBTAIN THE FOLLOWING (ALL SWITCHES OPEN UNLESS SPECIFIED) (NEGATIVE SUPPLY LEAD CONNECTED TO TB3-7 OR TB2-1):

1. WITH 27.0 VDC APPLIED AT TB2-2 AND S7 CLOSED, ADJUST R3 FOR READING ON M2 OF 100 mA.
2. WITH 30.0 VDC APPLIED AT TB2-2 AND S2 CLOSED, ADJUST R4 FOR READING ON M2 OF 2A.
3. WITH 20.0 VDC APPLIED AT TB3-2 AND S4 CLOSED, ADJUST R1 FOR READING ON M5 OF 250 mA.
4. WITH 20.0 VDC APPLIED AT TB3-2 AND S6 CLOSED, ADJUST R2 FOR READING ON M5 OF 125 mA.

FIGURE 1. Test fixture for the 1 zone power supply.

X-3885B

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APPENDIX

TEST METHOD 3

5 ZONE POWER SUPPLY

FUNCTIONAL TESTS

1. TEST APPARATUS

a.	Oscilloscope (Scope)	- Tektronix Model 5440 with Type 5A48 plug-in or equivalent.
b.	Voltmeter, dc (M3, M6)	- 3-1/2 digit or greater, minimum accuracy 0.25% of reading + 1 digit.
c.	Ammeter, DC (M5)	- 3-1/2 digit or greater, minimum accuracy 0.75% of reading + 1 digit.
d.	Voltmeter, ac (M1, M7)	- 3-1/2 digit or greater, minimum accuracy 1% of reading + 1 digit.
e.	Ammeter, ac (M4)	- 3-1/2 digit or greater, minimum accuracy 1.5% of reading + 1 digit.
f.	Ammeter, dc (M2)	- 3-1/2 digit or greater, minimum accuracy 1.5% of reading + 1 digit.
g.	Autotransformer (VR1)	- General Radio Model W20M or equivalent.
h.	SPST switches (S1-S8)	- Leviton Model 19916 or equivalent.
i.	Resistor Decade (R5)	- Clarostat Model 240C or equivalent.
j.	Resistor, 16 ohm (R1)	- Ohmite/Dividohm, 25 ohm, adjustable, 50 watts, P/N 0562 or equivalent.
k.	Resistor, 32 ohm (R2)	- Ohmite/Dividohm, 50 ohm, adjustable, 50 watts, P/N 0563 or equivalent.
l.	Resistor, 20 Kilohm (R6)	- Allen/Bradley/Type GB, 20K , ±5%, 1 watt or equivalent.
m.	Resistor, 180 ohm (R3)	- Ohmite/Dividohm, 200 ohm, adjustable, 12 watts, P/N 1014 or equivalent.
n.	Resistor, 7.5 ohm (R4)	- Clarostat, 10 ohm, adjustable, 225 watts, type VK-200-WA.
o.	Battery, 24V, 28A-H (B1)	- Eagle-Picher, CF 24V30 - no substitute.
p.	Test fixture	- (See figure 2).

NOTE: Power Supply, such as Hewlett-Packard Model 6291A or equivalent is required for calibrating test fixture resistors only.

2. TEST PREPARATION

2.1 When this test is being performed as part of the monitor cabinet functional test, either completely remove the power supply from the cabinet prior to performing the test procedure, or disconnect the wires going into terminals 2 and 3 of the signal module terminal board. Care should be taken to insure these wires do not make contact with any other wires.

3. TEST PROCEDURE

3.1 Output voltage levels.

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APPENDIX

a. Connect the power supply under test in the test fixture as shown in figure 2. Set power supply under test power switch in the OFF position and apply 115 Vac, 60 Hz line power to the autotransformer (VRI).
b. Open S1 through S8.
c. Adjust the VRI for 115 Vac on M1.
d. Close S3.
e. Place the power supply under test power switch in the ON position. Allow at least 15 minutes warm up before proceeding with test.
f. The following output voltage levels shall be observed:

Output	Location	Meter	Voltage
1	TB3-3 to TB3-5	M6	20 ±1 Vdc
2	TB2-4 to TB2-1	M3	See below
3	TB3-1 to TB3-6	M7	22 ±2 Vac

Output 2 shall be: 29 ±1.0 Vdc at 76 ±8 °F; 31 ±1.0 Vdc at -20 °F; 27 ±1.0 Vdc at +150 °F.

g. The output voltages must be within the above levels to be acceptable.

3.2 Line regulation.

a. Close S3, S4, and S5 and open S1, S2, S6, S7, and S8.
b. Place the power supply under test power switch in the ON position.
c. Adjust the VRI for 115 Vac reading on M1.
d. Observe the following output current levels:

Output	Location	Meter	Current
1	TB3-2 to TB3-7	M5	1.25 A ±5%
2	TB2-4 to TB2-1	M2	0 A
3	TB3-1 to TB3-7	M4	1.1 mA ac ±15%

e. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-3 to TB3-5	M6
2	TB2-4 to TB2-1	M3
3	TB3-1 to TB3-6	M7

f. Adjust the VRI for 125 Vac.
g. Repeat step e.
h. Adjust the VRI for 110 Vac.
i. Repeat step e.
j. For acceptable line regulation, the output voltage measurements taken in steps g. and i. shall not differ from those taken in step e., by more than the following:

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APPENDIX

1. Output 1 ±0.5%.
2. Output 2 ±1%.
3. Output 3 ±10%.

3.3 Load regulation.

a. Close S3 and S7 and open S1, S2, S4, S5, S6, and S8.
b. Adjust VR1 for 115 Vac as indicated on M1.
c. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-3 to TB3-5	M6
3	TB3-1 to TB3-6	M7

d. Close S4 and S5. Repeat the voltage readings as in step c. M5 shall indicate 1.25 ±.07 A and M4 shall indicate 1.100 ±0.170 mA ac.
e. Open S4 and close S6. Repeat the voltage readings as in step c. M5 shall indicate 0.62 ±.04 A and M4 shall indicate 1.100 ±0.170 mA ac.
f. For acceptable load regulation, the output voltage measurements taken in steps d. and e. shall not differ from those taken in step c, by more than the following:

1. Output 1 ±1%.
2. Output 3 ±5%.

3.4 Ripple and noise (periodic and random deviations).

a. Open S1, S2, S6, and S8 and close S3, S4, S5, and S7.
b. Measure the output ac P-P voltage at output 1 (TB3-3 to TB3-5) with the scope. The lead lengths from the scope to TB3 should be as short as possible.
c. For acceptable ripple and noise content, the ac P-P voltage shall be less than 400 mV.

3.5 Standby power.

NOTE: Steps a., b., c., m., n., and o. shall not be performed during burn-in, high temperature and low temperature tests.

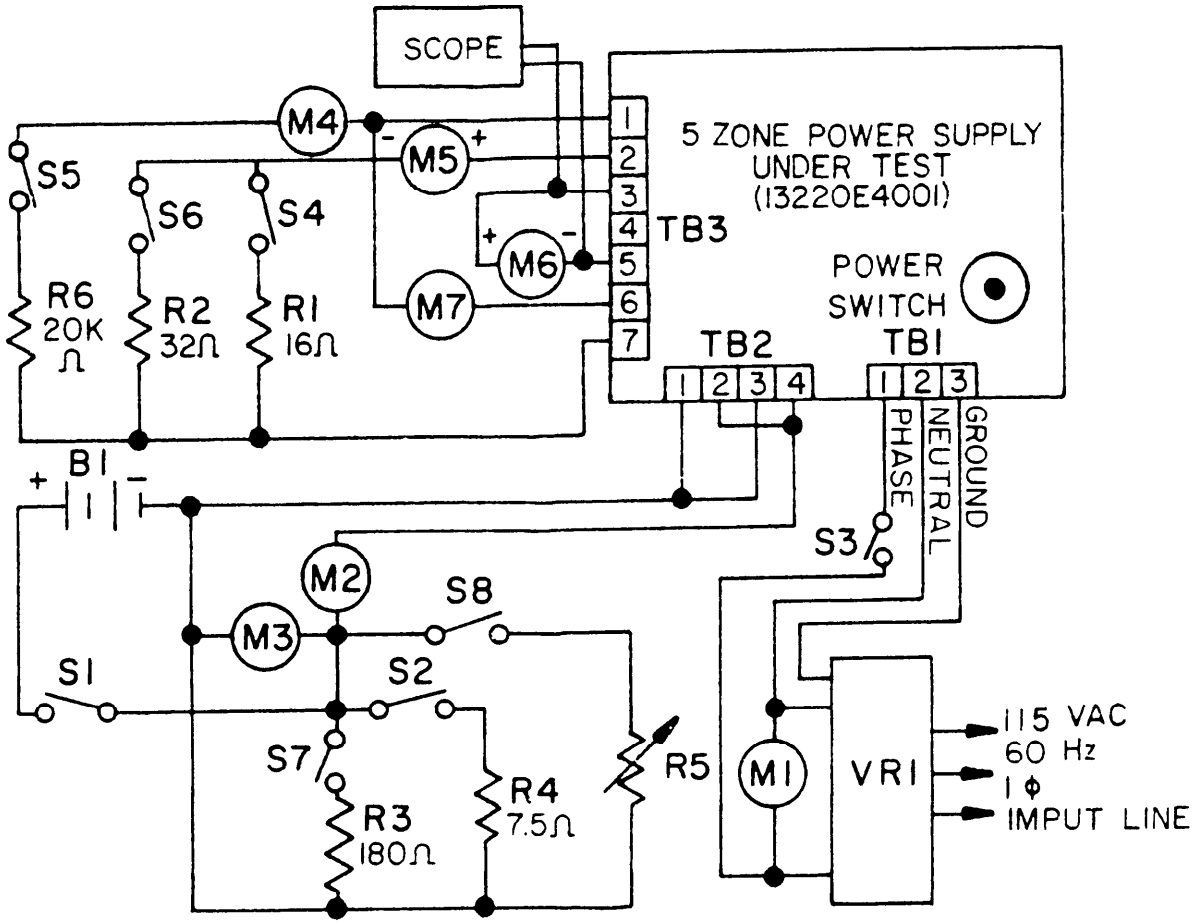
a. Close S1, S3, S5, and S6 and open S2, S4, S7, and S8.
b. Observe current on M2. Current shall be less than 50 mA. If not, wait until current is less than 50 mA before continuing. If current will not become less than 50 mA, replace batteries with fresh set.
c. When current measured in step b. above is less than 50 mA, observe voltage on M3. Voltage shall be 27.6 ±0.3 Vdc at room ambient temperature of 77 ±5 °F. When ambient temperature is outside this range during "room temperature" testing or at 86 °F humidity test, the voltage shall be in accordance with equation V=29.3 -.022T, where T is the temperature in degrees F.

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d. Close S3, S5, S6, and S7 and open S1, S2, S4 and S8.
e. Observe voltage on M3. Voltage shall be 27.0 ±0.5 Vdc at room ambient temperature of 77 ±5 °F, 29.6 ±0.6 Vdc at chamber temperature of -20 °F, and 25.8 ±0.6 Vdc at chamber temperature of 150 °F.
f. Remove scope probes from TB3 and place them across TB2-2 (positive) and TB2-1. Adjust vertical scale on scope to be able to be read from 15 Vdc to 33 Vdc with best feasible resolution.
g. Open S7 and close S2. The voltage on scope shall fluctuate between 29.5 ±3.5 and 17 ±2 Vdc. Proceed to step i. within 15 seconds.
h. Open S2 and close S7. Voltage on M3 shall be within 0.2 Vdc of voltage measured in step e.
i. Adjust R5 for 180 ohms, then open S7 and close S8. Voltage as read on M3 shall be as in step h.
j. Slowly reduce resistance of R5 while observing voltage on M3 and scope. Note voltage on M3 just before the voltage begins to fluctuate. Voltage just prior to fluctuation shall be 24.3 ±0.3 Vdc at temperature of 77 ±10 °F, and shall be no less than 23 Vdc at -20 °F and no greater than 25 Vdc at ±150 °F. Proceed to step k. within 15 seconds.
k. Increase resistance of R5 until voltage becomes steady. Voltage just prior to this change shall be 2 ±0.5 Vdc greater than that measured in step j.
l. Open S6 and S8 and close S1 and S4.
m. Open S3.
n. Observe voltage on meter M6 and current through meter M2. Voltage shall be 20 ±1 Vdc and current shall be 1.25 ±0.25 A.
o. Turn off monitor cabinet power, open all switches S1 through S8, and restore wiring to original configuration, if applicable.

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CALIBRATION PROCEDURE

USING REGULATED POWER SUPPLY IN PLACE OF 13220E4001 POWER SUPPLY, ADJUST VARIABLE RESISTORS SUCCESSIVELY TO OBTAIN THE FOLLOWING (ALL SWITCHES OPEN UNLESS SPECIFIED) (NEGATIVE SUPPLY LEAD CONNECTED TO TB3-7 OR TB2-1):

1. WITH 27.0 VDC APPLIED AT TB2-2 AND S7 CLOSED, ADJUST R3 FOR READING ON M2 OF 150 mA.
2. WITH 30.0 VDC APPLIED AT TB2-2 AND S2 CLOSED, ADJUST R4 FOR READING ON M2 OF 4A.
3. WITH 20.0 VDC APPLIED AT TB3-2, CLOSE S4 AND ADJUST RI FOR READING ON M5 OF 1.25 A.
4. WITH 20.0 VDC APPLIED AT TB3-2, CLOSE S6 AND ADJUST R2 FOR READING ON M5 OF 0.625A.

FIGURE 2. Test fixture for the 5 zone power supply.

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APPENDIX

TEST METHOD 4

25 ZONE POWER SUPPLY

FUNCTIONAL TESTS

1. TEST APPARATUS.

a.	Oscilloscope (scope)	- Tektronix Model 5440 with Type 5A48 plug-in or equivalent.
b.	Voltmeter, dc (M3, M6, M8)	- 3-1/2 digit or greater, minimum accuracy 0.25% of reading + 1 digit.
c.	Ammeter, dc (M5)	- 3-1/2 digit or greater, minimum accuracy 1.5% of reading + 1 digit.
d.	Voltmeter, ac (M1, M7)	- 3-1/2 digit or greater, minimum accuracy 1% of reading + 1 digit.
e.	Ammeter, ac (M4)	- 3-1/2 digit or greater, minimum accuracy 1.5% of reading + 1 digit.
f.	Ammeter, dc (M2)	- 3-1/2 digit or greater, minimum accuracy 1.5% of reading + 1 digit.
g.	Power Supply (PS1)	- Hewlett Packard Model 6267B or equivalent.
h.	Thermometer	- H-P, P/N HP2802A or equivalent.
i.	Display	- H-P, P/N HP34740A or equivalent.
j.	Probe	- H-P, P/N HP18640 PRT or equivalent.
k.	Heat Gun	- Ungar, P/N 6966C, or equivalent.
l.	Autotransformer (VR1)	- General Radio Model W20M or equivalent.
m.	SPST switches (S1-S10)	- Leviton Model 1991b or equivalent.
n.	Resistor Decade (R5)	- Clarostat Model 240C or equivalent.
o.	Resistor, 3.2 ohm (R1)	- Ohmite/Dividohm, 5 ohm, adjustable, 225 watts, P/N 1356 or equivalent.
p.	Resistor, 6.4 ohm (R2)	- Ohmite/Dividohm, 10 ohm, adjustable, 100 watts, P/N 0957 or equivalent.
q.	Resistor, 20 kilohm (R6) (K ohm)	- Allen-Bradley Type GB, 20K ohm, ±5%, 1 Watt or equivalent.
r.	Resistor, 72 ohm (R3)	- Ohmite/Dividohm, 100 ohm, adjustable, 25 watts, P/N 0368 or equivalent.
s.	Resistor, 3.75 ohm (R4A, R4B)	- Ohmite/Dividohm, 2 ohm, adjustable, 225 watts, P/N 1353 or equivalent.
t.	Battery, 24 V, 28A-H (B1, B2, B3)	- Eagle Picher CF 24V30, no substitute.
u.	Test fixture	- (See figure 3).

NOTE: Power supply such as Hewlett Packard Model 6267B is required for calibrating test fixture resistors.

2. TEST PREPARATION.

2.1 When this test is being performed as part of the monitor cabinet functional test, either completely remove the power supply from the cabinet prior to performing the test procedure, or disconnect the wires going into terminals 2 and 3 of the signal module terminal board. Care should be taken to insure these wires do not make contact with any other wires.

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APPENDIX

3. TEST PROCEDURE

3.1 Output voltage levels.

a. Connect the power supply under test in the test fixture as shown in figure 3. Set power supply under test power switch in the OFF position and apply 115 Vac, 60 Hz line power to the autotransformer (VRI).
b. Open S1 through S10.
c. Adjust the VRI for 115 Vac on M1.
d. Close S3.
e. Place the power supply under test power switch in the ON position Allow at least 15 minutes warm-up before proceeding with test.
f. The following output voltage levels shall be observed:

Output	Location	Meter	Voltage
1	TB3-3 to TB3-5	M6	20 ±1 Vdc
2	TB2-4 to TB2-3	M3	See below
3	TB3-1 to TB3-6	M7	22 ±2 Vac

Output 2 shall be: 29.5 ±0.5 Vdc at 76 +8 °F, 31 ±1.0 Vdc at -20 °F, and 27.5 ±1.0 Vdc at +150 °F.

g. The output voltages must be within the above levels to be acceptable.

3.2 Line regulation.

a. Close S3, S4, and S5 and open S1, S2, S6, S7, S8, S9, and S10.
b. Place the power supply under test power switch in the ON position.
c. Adjust the VRI for a 115 Vac reading on M1.
d. Observe the following output current levels:

Output	Location	Meter	Current
1	TB3-2 to TB3-7	M5	6.25 A ±5%
2	TB2-4 to TB2-5	M2	0 A
3	TB3-1 to TB3-7	M4	1.1 mA ac ±15%

e. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-3 to TB3-5	M6
2	TB2-4 to TB2-3	M3
3	TB3-1 to TB3-6	M7

f. Adjust the VRI for 125 Vac.
g. Repeat step e.
h. Adjust the VRI for 110 Vac.
i. Repeat step e.

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of table II to determine conformance to the requirements of this specifi-

j. For acceptable line regulation, the output voltage measurements taken in steps g. and i. shall not differ from those taken in step e. by more than the following:

1. Output 1 ±0.5%.
2. Output 2 ±1%.
3. Output 3 ±10%.

3.3 Load regulation.

a. Close S3 and S7 and open S1, S2, S4, S5, S6, S8, S9, and S10.
b. Adjust VR1 for 115 Vac as indicated on M1.
c. Observe and record the voltages at the following outputs:

Output	Location	Meter
1	TB3-3 to TB3-5	M6
3	TB3-1 to TB3-6	M7

d. Close S4 and S5. Repeat the voltage readings as in step c. M5 shall indicate 6.25 ±.32 A and M4 shall indicate 1.100 ±0.170 mA ac.
e. Open S4 and close S6. Repeat the voltage readings as in step c. M5 shall indicate 3.125 ±0.16A and M4 shall indicate 1.100 ±0.170 mA ac.
f. For acceptable load regulation, the output voltage measurements taken in steps d. and e. shall not differ from those taken in step c. by more than the following:

1. Output 1 ±1%.
2. Output 3 ±5%.

3.4 Ripple and noise (periodic and random deviations).

a. Open S1, S2, S6, S8, S9 and S10 and close S3, S4, S5, and S7.
b. Measure the output ac P-P voltage at output 1 (TB3-3 to TB3-5) with the scope. The lead lengths from the scope to TB3 should be as short as possible.
c. For acceptable ripple and noise content, the ac P-P voltage shall be less than 400 mV.

3.5 Standby power.

NOTE: Steps a., b., c., m., n., and o. shall not be performed during burn-in, high temperature, and low temperature tests.

a. Close S1, S3, S4, and S5 and open S2, S6, S7, S8, S9, and S10.
b. Observe current on M2. Current shall be less than 125 mA. If not, wait until current is less than 125 mA before continuing. If current will not become less than 125 mA, replace batteries with fresh set.

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c. When current measured in step b. above is less than 125 mA, observe voltage on M3. Voltage shall be 27.6 ±0.3 Vdc at room ambient temperature of 77 ±5 °F. When ambient temperature is outside this range during "room temperature" testing or at 86 °F humidity test, the voltage shall be in accordance with the equation V=29.3 -.022T, where T is the temperature in degrees F.
d. Close S3, S5, S6, and S7 and open S1, S2, S4, S8, S9, and S10.
e. Observe voltage on M3. Voltage shall be 27.0 ±0.5 Vdc at room ambient temperature of 77 ±5 °F, 29.6 ±0.6 Vdc at chamber temperature of -20 °F, and 25.8 ±0.6 Vdc at chamber temperature of 150 °F.
f. Remove scope probes from TB3 and place them across TB2-2 (positive) and TB2-1. Adjust vertical scale on scope to be able to read from 15 Vdc to 33 Vdc with best feasible resolution.
g. Open S7 and close S2. The voltage on scope shall fluctuate between 29.5 ±3.5 and 15 ±5 Vdc. Proceed to step h. within 10 seconds.
h. Open S2 and close S7. Voltage on M3 shall be within 0.2 Vdc of voltage measured in step e.
i. Adjust R5 for 72 ohms, then open S7 and close S8. Voltage as read on M3 shall be in step h.
j. Slowly reduce resistance of R5 while observing voltage on M3 and scope. Note voltage on M3 just before the voltage begins to fluctuate. Voltage just prior to fluctuation shall be 24.3 ±0.3 Vdc at temperature of 77 ±10 °F, and shall be no less than 23 Vdc at -20 °F and no greater than 25 Vdc at ±150 °F. Proceed to step k. within 15 seconds.
k. Increase resistance of R5 until voltage becomes steady. Voltage just prior to this change shall be 2 ±0.5 Vdc greater than that measured on step j.
l. Open S6 and S8 and close S1 and S4.
m. Open S3.
n. Observe voltage on meter M6 and current through meter M2. Voltage shall be 20 ±1 Vdc and current shall be 6.25 ±1.25 A.
o. Turn off monitor cabinet power, open all switches S1 through S10.

3.6 Thermal Cutoff

NOTES:
(1) Power supply cover is to be removed for this test.
(2) This test need not be performed at -20 °F, +150 °F and at 86 °F during humidity test.

a. Open S1, S2, S4, S5, S6, S7, S8, and S10. Close S3 and S9.
b. Connect a thermometer via a temperature sensing probe to the heat sink, in close proximity to the thermal switch (S2 in power supply under test).
c. Place power supply under test power switch in the ON position and adjust VR1 for 115 Vac on M1.
d. M3 will pulse between 29.5 ±3.5 Vdc and 20 ±5 Vdc.
e. Direct heat gun (heat mode) to blow on thermal switch of power supply.

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f. When the thermal switch activates, remove the heat gun and observe the following:

1. Thermometer shall indicate 150 °F ±10%.
2. M2 shall indicate zero (0) amps.
3. M3 shall indicate 28 ±2 Vdc.

g. Allow power supply to cool down to cause the thermal switch to reset (close). Observe the following:

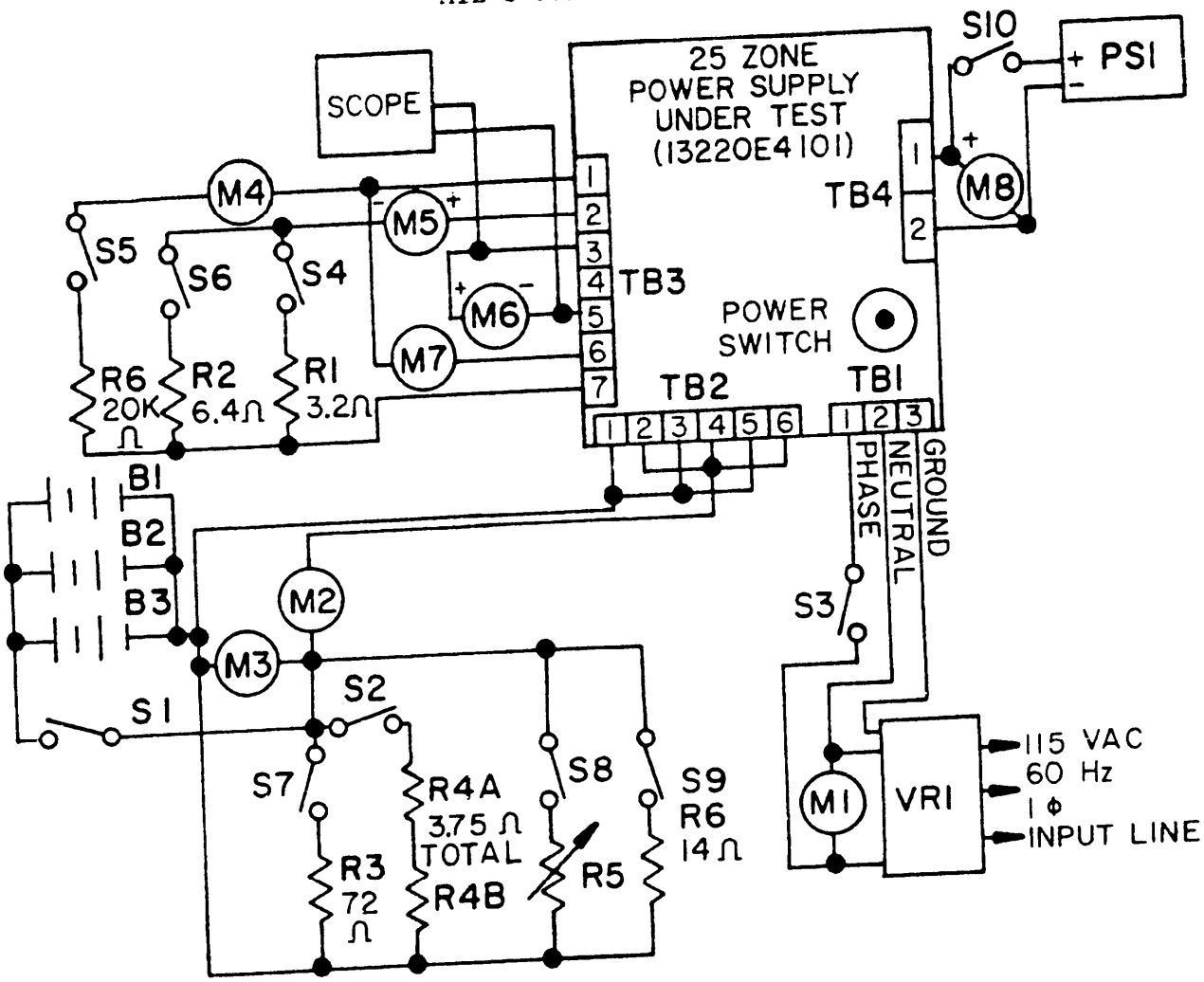
1. Thermometer shall indicate 130 °F ±10%.
2. M2 shall indicate as in step d. preceeding.
3. M3 shall indicate as in step d. preceeding.

h. Replace power supply cover on power supply under test.
i. Repeat steps a. through d. above.
j. When the thermal switch activates, M2 shall indicate 0 amps and M3 shall indicate 28 ±4 Vdc.
k. Allow power supply to cool down until thermal switch resets. The readings of step d. above shall repeat.
l. For acceptance, power supplier shall conform to above specifications.

3.7 Auxiliary dc power.

a. Open all switches on test fixture, except close S4, S5, and S10.
b. Adjust PSI to 25 ±1 V as indicated on M8.
c. M6 shall indicate 20 ±1 Vdc and M7 shall indicate 0 Vac.
d. For acceptance, the output voltage indications shall be within the above specified limits.

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CALIBRATION PROCEDURE

USING REGULATED POWER SUPPLY IN PLACE OF 13220E4101 POWER SUPPLY, ADJUST VARIABLE RESISTORS SUCCESSIVELY TO OBTAIN THE FOLLOWING (ALL SWITCHES OPEN UNLESS SPECIFIED) (NEGATIVE SUPPLY LEAD CONNECTED TO TB3-7 OR TB2-2):

1. WITH 27.0 VDC APPLIED AT TB2-2 AND S7 CLOSED, ADJUST R3 FOR READING ON M2 OF 375 mA.
2. WITH 30.0 VDC APPLIED AT TB2-2 AND S2 CLOSED, ADJUST R4 FOR READING ON M2 OF 8A.
3. WITH 20.0 VDC APPLIED AT TB3-2, CLOSE S4 AND ADJUST R1 FOR READING ON M5 OF 6.25A.
4. WITH 20.0 VDC APPLIED AT TB3-2, CLOSE S6 AND ADJUST R2 FOR READING ON M5 OF 3.125A.
5. WITH 30.0 VDC APPLIED AT TB2-2 AND S9 CLOSED, ADJUST R6 FOR READING ON M2 OF 2.14A.

FIGURE 3. Test fixture for the 25 zone power supply.

X-3887B

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TEST METHOD 5

SIGNAL MODULE

1. TEST APPARATUS

a.	Digital voltmeter	- Hewlett-Packard Model 34740A display, with 34702A multimeter or equivalent.
b.	Sound Level Meter	- B & K Type 2209 or equivalent.
c.	Test fixture-	- (see figure 4).

2. TEST PROCEDURE.

a. Close switch S1. Apply 20 Vdc to TB1-3. The signal module lamps shall be "on" and the audible signaling device (ASD) shall be silent.
b. Open switch S1. After 3 seconds, the signal module lamps shall flash and the ASD shall sound.
c. Place the ACKNOWLEDGE/TEST switch on signal module in the ACKNOWLEDGE position. The lamps shall stop flashing and extinguish, and the ASD shall be silent.
d. Place ACKNOWLEDGE/TEST switch in TEST position. The lamps shall light and remain lighted as long as switch is in TEST position. Release switch.
e. Close S1. The lamps shall flash and ASD shall sound. Place ACKNOWLEDGE/TEST switch in ACKNOWLEDGE position. The lamps shall stop flashing and remain on, and the ASD shall be silent.
f. Press switch S2. The ASD shall sound as long as switch is closed. Release S2, and the ASD shall be silent.
g. Measure the sound pressure level of the ASD at 10 feet directly in front of the signal module. The minimum acceptable sound pressure level shall be 60 dB.

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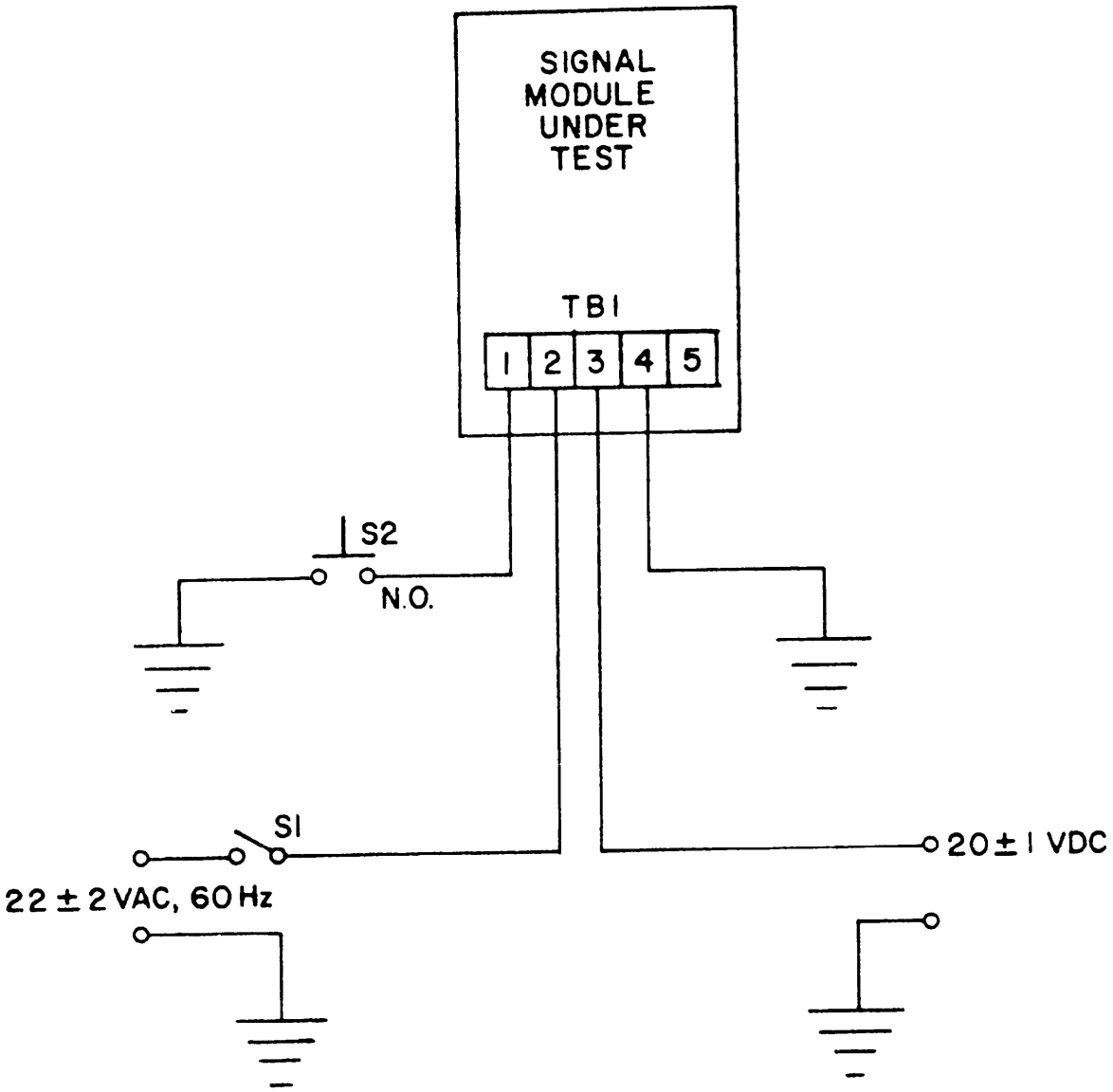


FIGURE 4. Test fixture for signal module.

X-3888

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(See Instructions – Reverse Side)

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MIL-C-0052876C(ME)	Cabinets, Monitor, CY-7359/FSS-9(V), CY-7360/FSS-9(V), and CY-7361/FSS-9(V)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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